



**NANYANG
TECHNOLOGICAL
UNIVERSITY**

CE4055

LAB:3

Understanding Smart Cards Communication

RED TEAM BLUE TEAM Game:

- Red Team (**Attacker**)
- Blue Team (**Defender**)
- Defender team has a password and acts like a password checker!!!
- Attacker team makes password queries and tries to crack the password.

RED TEAM BLUE TEAM Game:

BLUE TEAM (Defender)	RED TEAM (Attacker)
1. Create a 6 digit password. Each digit between 0 and 4	1. Need to guess the password in not more than 30 trials.
2. You are under an attack to recover the password.	2. Query the defender with password guesses.
3. An attacker can query with as many password guesses.	
HOW TO CHECK: 1. Check one digit at a time... If even a single digit mismatches, you abort the checking procedure and say FAIL. 2. But, you provide information on how many digits you checked. 3. If all the digit checks succeed, you say PASS.	
4. For a given query, you should provide following details: a) PASS or FAIL b) Number of digit checks performed before aborting the checking procedure.	

RED TEAM BLUE TEAM Game:

- Correct Password: 023414

Guessed Password	PASS/FAIL	Number of digits you checked before failure

- Send the table through email (Send all queries at once)
- Play the game with adjacent group as both defender and attacker!!!
- Who wins?? Attacker or Defender?

AGENDA:

- ❑ Hands-on session with Smart Cards and understanding how to communicate with Smart Cards.
- ❑ We will introduce the students to the hardware and software components in a basic smart card setup and trying to familiarize with them.
- ❑ We will work with two types of Smart cards:
 - ❑ CEPAS 2.0 card (Your ID card/EZLink Card)
 - ❑ MiFARE Ultralight Memory Card

What is a Smart Card?



- ❑ A smart card is a credit card sized plastic card with an embedded computer chip.
- ❑ Type 1: Microprocessor with internal memory
- ❑ Type 2: Memory chip with non-programmable logic.
- ❑ They can be programmed to accept, store and send data.

Difference between a Smart Card and a Magnetic Stripe card

☐ Magnetic Stripe card

- ☐ No chip
- ☐ Can only store data (magnetic stripes)
- ☐ Only contains **Meta data** (Card No, Expiry Date, CVV)
- ☐ E.g. North Spine Card
- ☐ <https://www.youtube.com/watch?v=Wbh9-gkSdrU>
- ☐ <https://www.youtube.com/watch?v=IITY2XukZek>
- ☐ **No Security!!!**

☐ Smart card (Contains Chip)

- ☐ More memory compared to magnetic stripe card.
- ☐ Has more powerful processing capabilities (can be used to perform operations such as data encryption, etc.)
- ☐ Uses Cryptography

Smart Card Readers

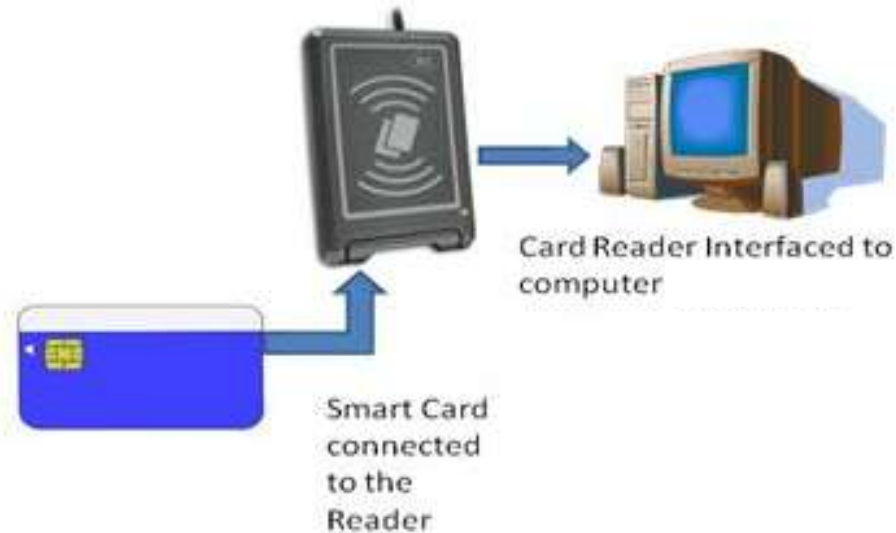


Computer based : needs to connect through USB or COM (Serial) ports

Dedicated Terminals: usually with a small screen, keyboard, etc.



A Basic Smart Card Working System



A Basic Smart Card Working System

- ❑ There are a number of communication interfaces here:
 - ❑ Smart Card to Smart card reader.
 - ❑ Smart Card Reader to Host-PC driver.
 - ❑ Host-PC software driver to User Application.

A Basic Smart Card Working System

- ☐ Back in the days, all the components involved in this setup were all **proprietary**.
- ☐ For example,
 - ☐ Samsung phones could only talk with Samsung Cell phone towers and Samsung cell phones.
 - ☐ Samsung will have its own OS.
 - ☐ Samsung will have to develop its own Whatsapp Application or Temple run Application.
- ☐ It would be a real pain for all the Mobile phone companies and also its customers right?
- ☐ This was the case with Smart cards back in the days.

Standardizing everything!!!

- ❑ Massive effort from industry for standardization!!!
 - ❑ Microsoft, Motorola, NXP and others.
- ❑ Today, a lot of standards exist for each of the interfaces present in a typical smart card environment.
- ❑ Smart-cards: **ISO/IEC 7816, ISO/IEC 14443**
- ❑ How Smart card readers talk to Host PC: **PC/SC standard (APIs)**
- ❑ Even Smart Cards have operating systems (**Chip Operating System**)
- ❑ Can host **multiple applications** on smart card (like a computer)

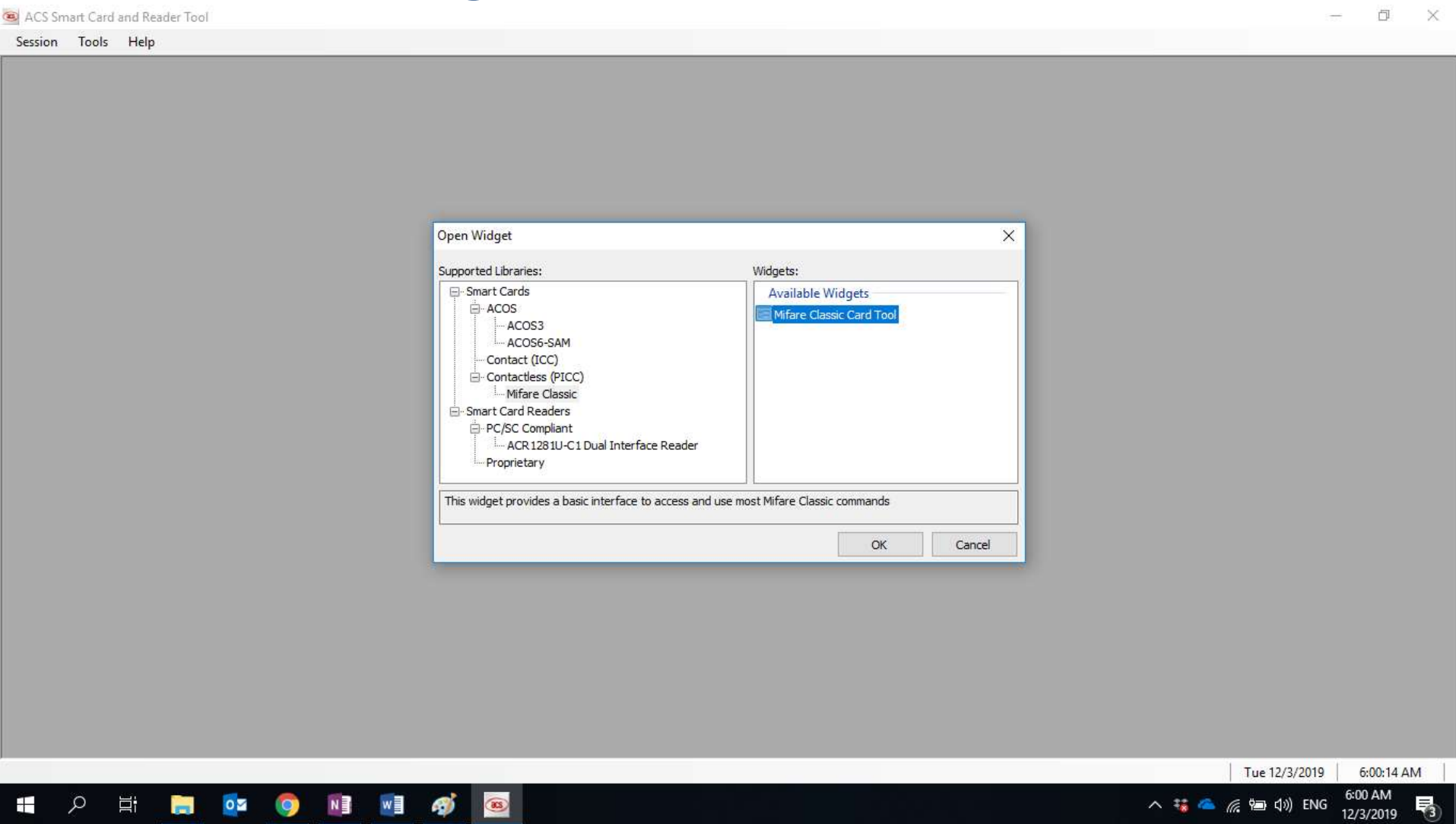
ACS Smart Card Reader Tool

- ☐ Talk to Smart Card Reader through PC
- ☐ Uses PCSC Driver (Similar to driver for mouse/keyboard)
- ☐ Sends commands to the smart card reader.
- ☐ The smart card reader interprets the commands and communicates with the smart card.
- ☐ Sends back response from smart card back to Host PC

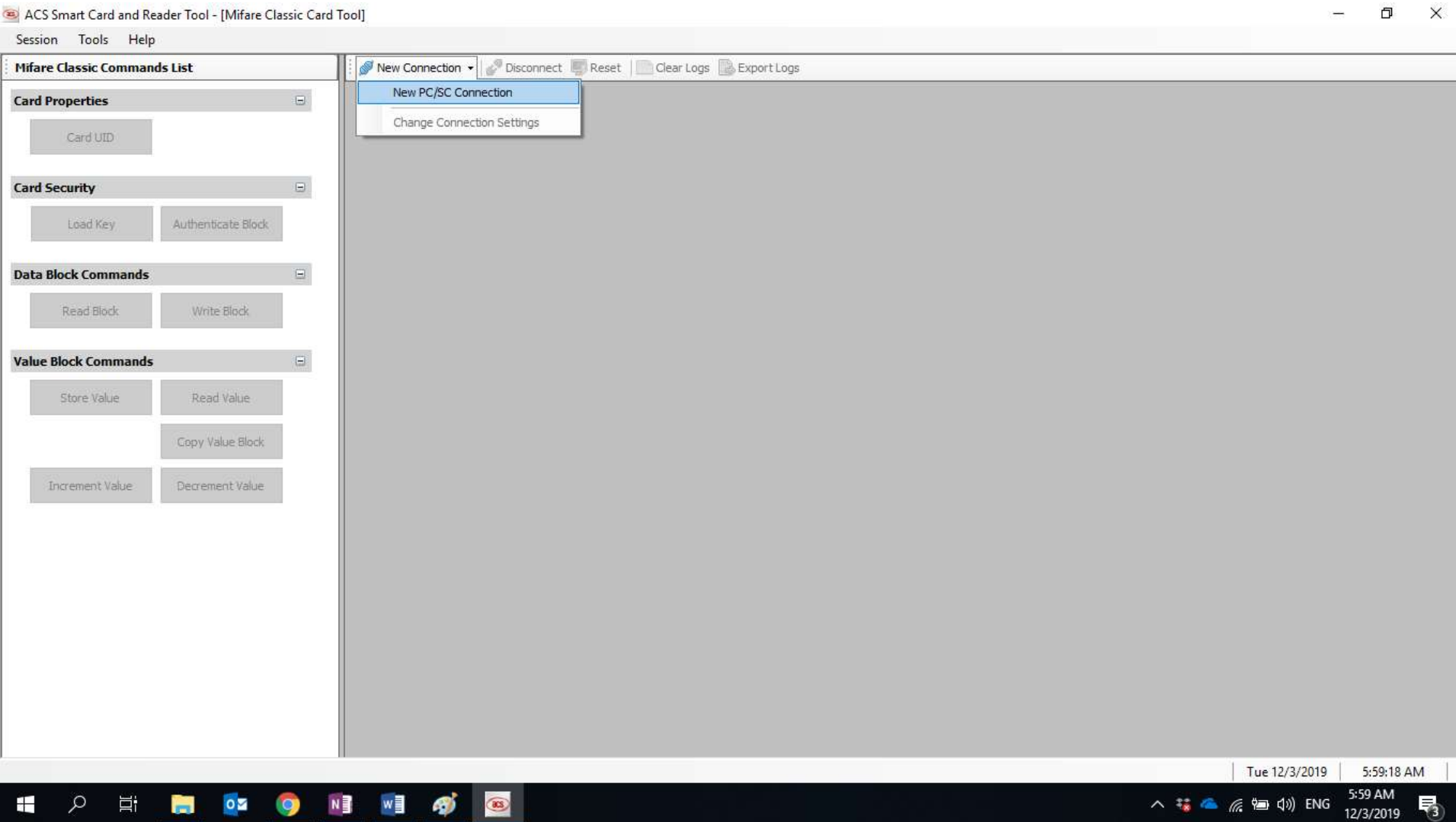
MiFARE Ultralight EV1:

- ❑ Contactless Smart Card
- ❑ Where is it used?
 - ❑ Single Journey Ticket in Public Transit
 - ❑ Limited Use Tickets in Public Transit
 - ❑ Day Pass for events
- ❑ Memory Card - With Some functionalities
- ❑ It has about **64** bytes of memory and has **48** bytes for storing data.
- ❑ It can also be **password authenticated**, but for our lab session, we use it as free-access cards.

Using ACS Smart Card and Reader Tool for MiFARE Ultralight Card



Establishing Connection with Smart Card Reader



Selecting the interface of the Smart Card Reader

ACS Smart Card and Reader Tool - [Mifare Classic Card Tool]

Session Tools Help

Mifare Classic Commands List

Card Properties

Card UID

Card Security

Load Key Authenticate Block

Data Block Commands

Read Block Write Block

Value Block Commands

Store Value Read Value

Copy Value Block

Increment Value Decrement Value

New Connection Disconnect Reset Clear Logs Export Logs

Connect to a PC/SC-compliant Contactless Smartcard Reader

Reader: ACS ACR1281 1S Dual Reader PICC 0
ACS ACR1281 1S Dual Reader ICC 0
ACS ACR1281 1S Dual Reader PICC 0
Connect: ACS ACR1281 1S Dual Reader SAM 0

Share Modes
☒ Shared
☐ Exclusive

Protocol
Mifare Classic cards uses
T=CL protocol

Connect Cancel

Tue 12/3/2019 6:01:01 AM

Windows taskbar icons: Start, Search, Task View, File Explorer, Edge, Chrome, Word, PowerPoint, ACS Smart Card and Reader Tool.

ANSWER TO RESET

ACS Smart Card and Reader Tool - [Mifare Classic Card Tool]

Session Tools Help

Mifare Classic Commands List

Card Properties

Card UID

Card Security

Load Key Authenticate Block

Data Block Commands

Read Block Write Block

Value Block Commands

Store Value Read Value

Copy Value Block

Increment Value Decrement Value

ACS ACR1281 1S Dual Reader PICC 0

New Connection Disconnect Reset Clear Logs Export Logs

Connected to ACS ACR1281 1S Dual Reader PICC 0

ATR: 3B 8F 80 01 80 4F 0C A0 00 00 03 06 03 00 03 00 00 00 68

← Answer to Reset (ATR)

- ☐ Answer to Reset is the first response on power up from the card.
- ☐ It does not uniquely identify the card...
- ☐ ATR is a way of saying “Hello” to the card reader... And using the type of “Hello”, the reader will know what language he needs to use.

Reader ACS ACR1281 1S Dual Reader PICC 0 Share Mode Shared Protocol T=CL

Tue 12/3/2019 6:01:24 AM

6:01 AM 12/3/2019

ANSWER TO RESET

4.2.2. ATR format for ISO 14443 Part 3 PICCs.

Byte	Value (Hex)	Designation	Description
0	3B _H	Initial Header	
1	8N _H	T0	Higher nibble 8 means: no TA1, TB1, TC1 only TD1 is following. Lower nibble N is the number of historical bytes (HistByte 0 to HistByte N-1)
2	80 _H	TD1	Higher nibble 8 means: no TA2, TB2, TC2 only TD2 is following. Lower nibble 0 means T = 0
3	01 _H	TD2	Higher nibble 0 means no TA3, TB3, TC3, TD3 following. Lower nibble 1 means T = 1
4 To 3+N	80 _H	T1	Category indicator byte, 80 means A status indicator may be present in an optional COMPACT-TLV data object
	4F _H	Tk	Application identifier Presence Indicator
	0C _H		Length
	RID		Registered Application Provider Identifier (RID) # A0 00 00 03 06
	88		Byte for standard
	C0... C1		Bytes for card name
	00 00 00 00 _H	RFU	RFU # 00 00 00 00
4+N	UU	TCK	Exclusive-oring of all the bytes T0 to Tk

Table 4: ISO 14443 Part 3 ATR Format

ANSWER TO RESET

Example:

ATR for Mifare 1K = {3B 8F 80 01 80 4F 0C A0 00 00 03 06 03 00 01 00 00 00 00 6A}

ATR											
Initial Header	T0	TD1	TD2	T1	Tk	Length	RID	Standard	Card Name	RFU	TCK
3B _H	8F _H	80 _H	01 _H	80 _H	4F _H	0C _H	A0 00 00 03 06 _H	03 _H	00 _H 01 _H	00 00 00 00 _H	6A _H

Where:

Length (YY) = 0C
 RID = A0 00 00 03 06 (PC/SC Workgroup)
 Standard (SS) = 03 (ISO 14443A, Part 3)
 Card Name (C0 ... C1) =
 [00 01] (Mifare 1K)
 [00 02] (Mifare 4K)
 [00 03] (Mifare Ultralight)
 [00 26] (Mifare Mini)
 [FF 28] JCOP 30
 FF SAK undefined tags

From the ATR, try to identify what card you are working with...

Memory Map of Mifare Ultralight:

Byte Number	0	1	2	3	Page
Serial Number	SN0	SN1	SN2	BCC0	0
Serial Number	SN3	SN4	SN5	SN6	1
Internal / Lock	BCC1	Internal	Lock0	Lock1	2
OTP	OPT0	OPT1	OTP2	OTP3	3
Data read/write	Data0	Data1	Data2	Data3	4
Data read/write	Data4	Data5	Data6	Data7	5
Data read/write	Data8	Data9	Data10	Data11	6
Data read/write	Data12	Data13	Data14	Data15	7
Data read/write	Data16	Data17	Data18	Data19	8
Data read/write	Data20	Data21	Data22	Data23	9
Data read/write	Data24	Data25	Data26	Data27	10
Data read/write	Data28	Data29	Data30	Data31	11
Data read/write	Data32	Data33	Data34	Data35	12
Data read/write	Data36	Data37	Data38	Data39	13
Data read/write	Data40	Data41	Data42	Data43	14
Data read/write	Data44	Data45	Data46	Data47	15

Table 8: Mifare Ultralight Memory Map

Serial Number:

SN0, SN1, SN2, SN3, SN4, SN5, SN6

Block Check Character (BCC):

BCC0: $CT \wedge SN0 \wedge SN1 \wedge SN2$

BCC1: $SN3 \wedge SN4 \wedge SN5 \wedge SN6$

512 bits
Or
64 bytes

Memory Map of Mifare Ultralight:

Byte Number	0	1	2	3	Page
Serial Number	SN0	SN1	SN2	BCC0	0
Serial Number	SN3	SN4	SN5	SN6	1
Internal / Lock	BCC1	Internal	Lock0	Lock1	2
OTP	OPT0	OPT1	OTP2	OTP3	3
Data read/write	Data0	Data1	Data2	Data3	4
Data read/write	Data4	Data5	Data6	Data7	5
Data read/write	Data8	Data9	Data10	Data11	6
Data read/write	Data12	Data13	Data14	Data15	7
Data read/write	Data16	Data17	Data18	Data19	8
Data read/write	Data20	Data21	Data22	Data23	9
Data read/write	Data24	Data25	Data26	Data27	10
Data read/write	Data28	Data29	Data30	Data31	11
Data read/write	Data32	Data33	Data34	Data35	12
Data read/write	Data36	Data37	Data38	Data39	13
Data read/write	Data40	Data41	Data42	Data43	14
Data read/write	Data44	Data45	Data46	Data47	15

512 bits
Or
64 bytes

There are about 48 user programmable bytes.

They can be read and written any number of times and does not have any access control.

Table 8: Mifare Ultralight Memory Map

Read Memory:

ACS Smart Card and Reader Tool - [Mifare Classic Card Tool]

Session Tools Help

Mifare Classic Commands List

- Card Properties**
 - Card UID
- Card Security**
 - Load Key
 - Authenticate Block
- Data Block Commands**
 - Read Block
 - Write Block
- Value Block Commands**
 - Store Value
 - Read Value
 - Copy Value Block
 - Increment Value
 - Decrement Value

New Connection Disconnect Reset Clear Logs Export Logs

ACS ACR1281 1S Dual Reader PICC 0

Read Block

Block To Read: 00

Length : 10

Show in ASCII: ☐

Read Cancel

Input Page Number and length of bytes to read...

Reader ACS ACR1281 1S Dual Reader PICC 0 Share Mode Shared Protocol T=CL

Tue 12/3/2019 8:13:58 AM

8:13 AM 12/3/2019 ENG

Read Memory:

ACS Smart Card and Reader Tool - [Mifare Classic Card Tool]

Session Tools Help

Mifare Classic Commands List

Card Properties

Card UID

Card Security

Load Key Authenticate Block

Data Block Commands

Read Block Write Block

Value Block Commands

Store Value Read Value

Copy Value Block

Increment Value Decrement Value

ACS ACR1281 1S Dual Reader PICC 0

New Connection Disconnect Reset Clear Logs Export Logs

> Read Block
> FF B0 00 00 10
< 04 CD AB EA C2 37 30 80 45 48 00 00 00 00 00
< 90 00

← Bytes read from memory

Data: 04 CD AB EA C2 37 30 80 45 48 00 00 00 00 00

Try to Read 4 bytes from page 4

Reader ACS ACR1281 1S Dual Reader PICC 0 Share Mode Shared Protocol T=CL

Tue 12/3/2019 8:18:17 AM

8:18 AM 12/3/2019

Read Memory:

ACS Smart Card and Reader Tool - [Mifare Classic Card Tool]

Session Tools Help

Mifare Classic Commands List

Card Properties

Card UID

Card Security

Load Key Authenticate Block

Data Block Commands

Read Block Write Block

Value Block Commands

Store Value Read Value

Copy Value Block

Increment Value Decrement Value

ACS ACR1281 1S Dual Reader PICC 0

New Connection Disconnect Reset Clear Logs Export Logs

> Read Block
> FF B0 00 04 04
< 00 00 00 00
< 90 00

← Observe the command structure...

Data: 00 00 00 00

Command	Class	INS	P1	P2	Le
Read Binary Blocks	FF _H	B0 _H	00 _H	Block Number	Number of Bytes to Read

Reader ACS ACR1281 1S Dual Reader PICC 0 Share Mode Shared Protocol T=CL

Tue 12/3/2019 8:24:10 AM

8:24 AM 12/3/2019

Update Memory:

ACS Smart Card and Reader Tool - [Mifare Classic Card Tool]

Session Tools Help

Mifare Classic Commands List

- Card Properties
 - Card UID
- Card Security
 - Load Key
 - Authenticate Block
- Data Block Commands
 - Read Block
 - Write Block**
- Value Block Commands
 - Store Value
 - Read Value
 - Copy Value Block
 - Increment Value
 - Decrement Value

ACS ACR1281 1S Dual Reader PICC 0

New Connection Disconnect Reset Clear Logs Export Logs

```
> Read Block
> FF B0 00 04 04
< 00 00 00 00
< 90 00
Data: 00 00 00 00
```

Write Block

Target Block: 04 Length Of Data: 04

Data To Write

10	11	12	13												
----	----	----	----	--	--	--	--	--	--	--	--	--	--	--	--

Write Cancel

Reader ACS ACR1281 1S Dual Reader PICC 0 Share Mode Shared Protocol T=CL

Tue 12/3/2019 8:31:54 AM

Update Memory:

ACS Smart Card and Reader Tool - [Mifare Classic Card Tool]

Session Tools Help

Mifare Classic Commands List

Card Properties

Card UID

Card Security

Load Key Authenticate Block

Data Block Commands

Read Block Write Block

Value Block Commands

Store Value Read Value

Copy Value Block

Increment Value Decrement Value

ACS ACR1281 1S Dual Reader PICC 0

New Connection Disconnect Reset Clear Logs Export Logs

> Read Block
> FF B0 00 04 04
< 00 00 00 00
< 90 00
Data: 00 00 00 00

> Write Block
> FF D6 00 04 04 10 11 12 13
< 90 00
Write Block OK

← Response for Write Command...

Perform read to the same memory block to see if write was successful...

Reader ACS ACR1281 1S Dual Reader PICC 0 Share Mode Shared Protocol T=CL

Tue 12/3/2019 8:33:07 AM

8:33 AM 12/3/2019

Update Memory:

ACS Smart Card and Reader Tool - [Mifare Classic Card Tool]

Session Tools Help

Mifare Classic Commands List

Card Properties

Card UID

Card Security

Load Key Authenticate Block

Data Block Commands

Read Block Write Block

Value Block Commands

Store Value Read Value

Copy Value Block

Increment Value Decrement Value

ACS ACR1281 1S Dual Reader PICC 0

New Connection Disconnect Reset Clear Logs Export Logs

> Read Block
> FF B0 00 04 04
< 00 00 00 00
< 90 00

← Old Data

Data: 00 00 00 00

> Write Block
> FF D6 00 04 04 10 11 12 13
< 90 00

Write Block OK

> Read Block
> FF B0 00 04 04
< 10 11 12 13
< 90 00

← New Data

Data: 10 11 12 13

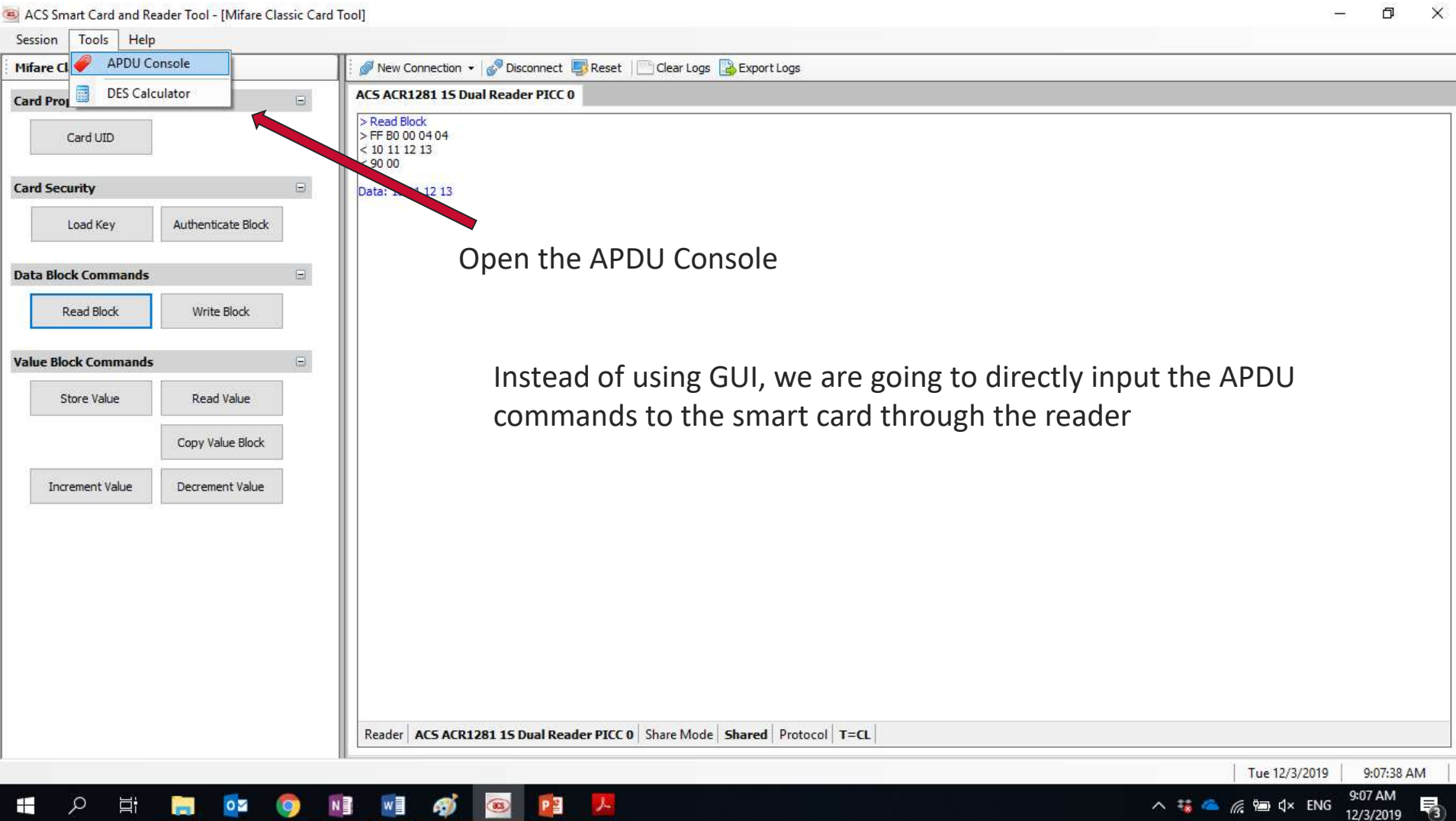
Command	Class	INS	P1	P2	Le	Data In
Update Binary Blocks	FF _H	D6 _H	00 _H	Block Number	Number of Bytes to Update	Block Data (Multiple of 16 Bytes)

Reader ACS ACR1281 1S Dual Reader PICC 0 Share Mode Shared Protocol T=CL

Tue 12/3/2019 8:34:48 AM

8:34 AM 12/3/2019

Sending Commands through APDU Console



The screenshot displays the 'ACS Smart Card and Reader Tool - [Mifare Classic Card Tool]' application. The 'Tools' menu is open, and the 'APDU Console' option is highlighted with a red arrow. The main window shows a connection to 'ACS ACR1281 1S Dual Reader PICC 0' with a log of commands and data. The left sidebar contains sections for 'Card UID', 'Card Security', 'Data Block Commands', and 'Value Block Commands'. The Windows taskbar at the bottom shows the date as Tuesday, December 3, 2019, at 9:07 AM.

Session Tools Help

Mifare Classic Card Tool

APDU Console

DES Calculator

Card UID

Card Security

Load Key Authenticate Block

Data Block Commands

Read Block Write Block

Value Block Commands

Store Value Read Value

Copy Value Block

Increment Value Decrement Value

ACS ACR1281 1S Dual Reader PICC 0

> Read Block
> FF B0 00 04 04
< 10 11 12 13
< 90 00
Data: 10 11 12 13

Reader ACS ACR1281 1S Dual Reader PICC 0 Share Mode Shared Protocol T=CL

Tue 12/3/2019 9:07:38 AM

9:07 AM 12/3/2019

Open the APDU Console

Instead of using GUI, we are going to directly input the APDU commands to the smart card through the reader

Sending Commands through APDU Console

ACS Smart Card and Reader Tool - [Mifare Classic Card Tool]

Session Tools Help

Mifare Classic Commands List

Card Properties

Card UID

Card Security

Load Key Authenticate Block

Data Block Commands

Read Block Write Block

Value Block Commands

Store Value Read Value

Copy Value Block

Increment Value Decrement Value

ACS ACR1281 1S Dual Reader PICC 0

New Connection Disconnect Reset Clear Logs Export Logs

> Read Block
> FF B0 00 04 04
< 10 11 12 13
< 90 00

Data: 10 11 12 13

> APDU
> FF B0 00 04 04
< 10 11 12 13
< 90 00

Get Response here!!!

APDU Command Console

Active Reader: **ACS ACR1281 1S Dual Reader PICC 0**

APDU Command

CLA	INS	P1	P2	Lc	Le
FF	B0	00	04	00	04

Input Command here!!!

Data In Parameters

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
► +00																
+10																
+20																

Data Out Result

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
► +00	10	11	12	13												
+10																
+20																

SW1 SW2
90 00

Transmit Clear Close

0 - The operation completed successfully

Reader **ACS ACR1281 1S Dual Reader PICC 0** Share Mode **Shared** Protocol **T=CL**

Tue 12/3/2019 9:12:38 AM

Windows Taskbar: File Explorer, Edge, Word, PowerPoint, PDF Reader, ACS Tool, etc.

Memory Map of Mifare Ultralight:

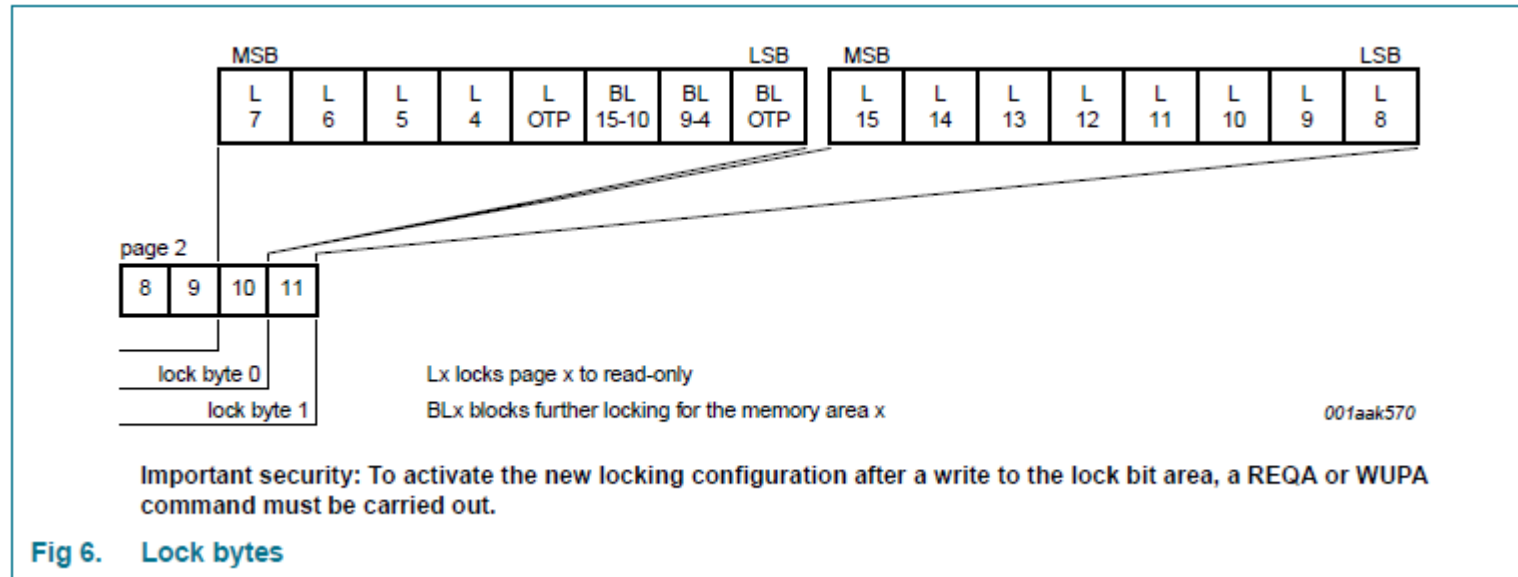
Byte Number	0	1	2	3	Page
Serial Number	SN0	SN1	SN2	BCC0	0
Serial Number	SN3	SN4	SN5	SN6	1
Internal / Lock	BCC1	Internal	Lock0	Lock1	2
OTP	OPT0	OPT1	OTP2	OTP3	3
Data read/write	Data0	Data1	Data2	Data3	4
Data read/write	Data4	Data5	Data6	Data7	5
Data read/write	Data8	Data9	Data10	Data11	6
Data read/write	Data12	Data13	Data14	Data15	7
Data read/write	Data16	Data17	Data18	Data19	8
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Data read/write	Data24	Data25	Data26	Data27	10
Data read/write	Data28	Data29	Data30	Data31	11
Data read/write	Data32	Data33	Data34	Data35	12
Data read/write	Data36	Data37	Data38	Data39	13
Data read/write	Data40	Data41	Data42	Data43	14
Data read/write	Data44	Data45	Data46	Data47	15

There are special bits called
Lock Bits
OTP Bits

512 bits
Or
64 bytes

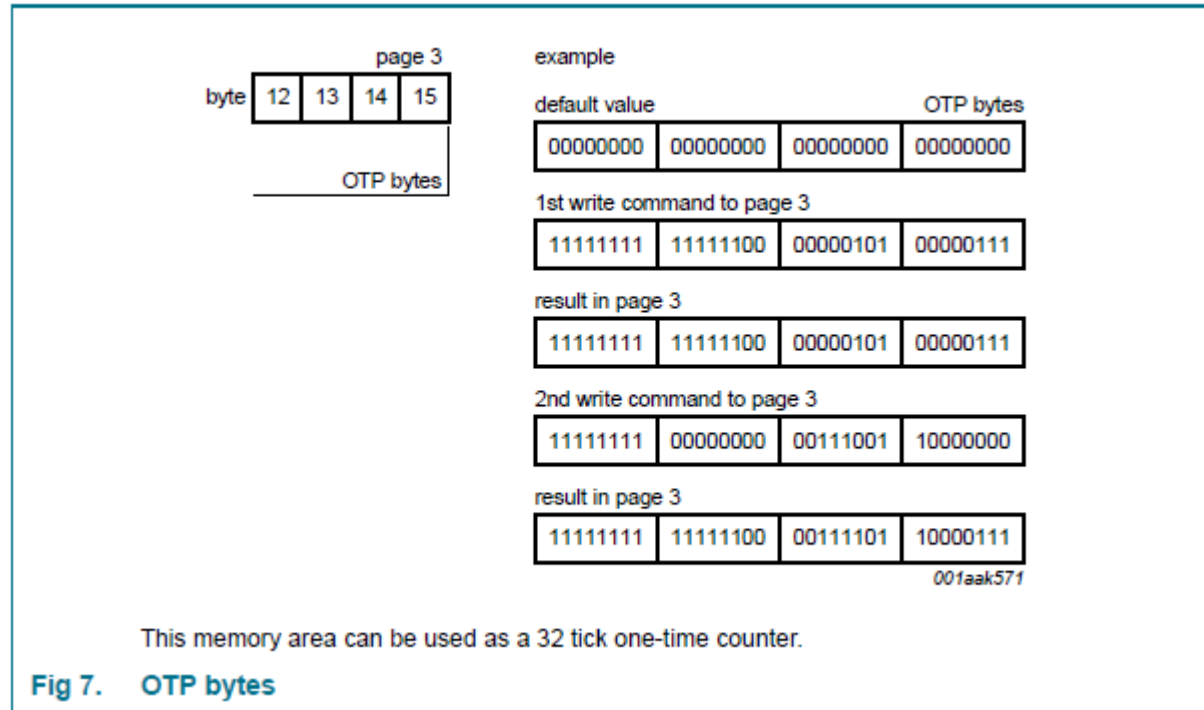
Table 8: Mifare Ultralight Memory Map

Lock bits of Mifare Ultralight:



Can be used to individually lock pages in the memory.
Beware: These lock bits are 0 by default and once-programmed, the corresponding pages can never be written to.

OTP bits of Mifare Ultralight:



One-Time Programmable bytes in the memory.

Beware: These bits are 0 by default and once a bit is programmed to 1, it can never be reverted back to 0.

ACTIVITY:

- ☐ Fill in the pages (05 and 06) of your MiFARE card with data in the following format:
 - ☐ The first byte can be chosen randomly.
 - ☐ The second byte is the XOR of the first byte with 0xFF.
 - ☐ The third byte is the XNOR of the first two bytes.
 - ☐ The fourth byte is simply the even-parity bit of the first three bytes.
 - ☐ But, keep in mind, the XOR of the first byte of the two pages should be 0xCC.

- ☐ Exchange your card with your neighbouring group and get the validity of your data format verified by them.

ACTIVITY:

CEPAS 2.0 Card

- ❑ **What CEPAS Stands for?**
 - ❑ Specification for **C**ontactless **E**-**P**urse **A**pplication
- ❑ Singaporean standard for electronic money card
- ❑ Singaporean vision of using single smart card for all services.
- ❑ Provides details on:
 - ❑ How to communicate with the card.
 - ❑ How the data is managed inside the card.
 - ❑ How to perform credit and debit (What kind of crypto to use?)
- ❑ NETS and EZ-Link cards have to adhere to CEPAS specifications in Singapore.

CEPAS 2.0 Card

- ❑ Credit and Debit functions are protected by crypto!!!
- ❑ What can we do?
 - ❑ Read purse balance from the purse file
 - ❑ Read transaction log.
- ❑ We use APDU Plus tool to communicate with CEPAS card

APDU Plus

- ☐ Send plain APDU commands to the smart card through the smart card reader.
- ☐ We will use this tool to read purse balance and transaction log from a CEPAS 2.0 card.

APDU Plus

APDU Plus

File Script Spawn About...

Identifying the different readers connected to the PC

Readers

Reader Name	Card Status	ATR
ACS ACR1281 1S Dual Reader ICC 0	Removed	
ACS ACR1281 1S Dual Reader PICC 0	Inserted	3B 8F 80 01 80 4F 0C A0 00 00 03 0...
ACS ACR1281 1S Dual Reader SAM 0	Removed	

Please select a reader from the above list.

Share Mode: Protocol: Disposition:

Refresh

Auto-refresh ☐

Interval (sec):

Connect

Disconnect

Commands

Script [no file loaded]

APDU: Transmit

Command: Transmit

Stress: ☐ Interval (ms): Pause

Input actual APDU to be transmitted here

Output

Output to log: ☒ Clear Log View Log

Output Log can be seen here...

Clear

```
ScardListReaders      : SCARD_S_SUCCESS
Number of Readers     : 3
ACS ACR1281 1S Dual Reader ICC 0; Card Removed
ACS ACR1281 1S Dual Reader PICC 0; Card Inserted (ATR: 3B 8F 80 01 80 4F 0C A0 00 00 03 06 03 0)
ACS ACR1281 1S Dual Reader SAM 0; Card Removed
```

Exit

Connect to
the Smart
Card reader

READING PURSE BALANCE OF CEPAS CARD

- ☐ Purse balance is stored in a Purse File.
- ☐ Input the APDU “90 32 03 00 00” to read purse balance.

CLA	'90' (proprietary class)
INS	'32'
P1	Encodes a short EF identifier of the File ID of the Purse File - Pf
P2	RFU
L_c field	'00' (no authentication), '01' (Read Transaction Log) or '0A' (with authentication)
Data field	See Read Purse Command Data
L_e field	Number of byte to be read

- ☐ Each purse file is identified by unique ID. Our ID is??
- ☐ Purse balance is specified by 3 byte signed integer. Why signed??

STRUCTURE OF PURSE BALANCE RECORD

Data object	Length	Security
CEPAS version	1 byte	None
Purse status	1 byte	None
Purse balance	3 bytes	None
AutoLoad amount	3 bytes	None
CAN	8 bytes	None
CSN	8 bytes	None
Purse expiry date (Julian Date)	2 bytes	None
Purse creation date (Julian Date)	2 bytes	None
Last credit transaction TRP	4 bytes	None
Last credit transaction header	8 bytes	None
No. of records in transaction logfile	1 byte	None
Issuer-specific data length	1 byte	None
Last transaction TRP	4 bytes	None
Last transaction record	16 bytes	None
Issuer-specific data	X bytes	None

Data object	Length	Security
Last transaction debit options byte	1 byte	None
Last transaction signed certificate	8 bytes	Triple-DES CBC
Counter data	8 bytes	
ISO/IEC 14443-3 (Annex B) CRC_B	2 bytes	None

READ TRANSACTION LOG OF CEPAS CARD

- ☐ Transaction log records are also sometimes stored along with Purse File in the card.
- ☐ Best Data types to store Transaction log. FIFO or LIFO??
- ☐ When is the transaction log updated?
- ☐ Input the APDU “90 32 03 00 01 00 00” to read transaction log record.
- ☐ In this case, is the transaction log record stored in the same file as the Purse file??

Transaction log record format			
Transaction header			Transaction user data
Transaction type	Transaction amount	Transaction DateTime	
1 byte	3 bytes	4 bytes	8 bytes

ACTIVITY:

- ☐ Read the Purse Balance and Transaction log of your card using an Android App for reading Smart Card Readers (SingCARD and many more are available...)
- ☐ Note it down and pass the card to your neighbouring group.
- ☐ Manually read the purse balance and transaction record of the shared card and check with the group if the information is correct.